John Francis Burkhart

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Education

Northern Arizona University *Flagstaff, AZ*
B.S. Mathematics, Summa Sum Laude. GPA: 4.0/4.0 *Aug. 2018 - May 2022*

Publications

1. David A Rasmussen, Madeline G Bursell, **Frank Burkhart**, Optimizing genomic sampling for demographic and epidemiological inference with Markov decision processes, *Genetics*, 2025;, iyaf244, https://doi.org/10.1093/genetics/iyaf244
2. Enumerating signed permutations by reversal distance. F. Awik, **F. Burkhart**, H. Denoncourt, D.C. Ernst, T. Rosenberg, and A. Stewart (in preparation).

Research Experience

North Carolina State University *Raleigh, NC*

Graduate Research Assistant *Aug. 2023 - April 2024*

- Co-authored and published paper on optimizing genome sampling for demographic and epidemiological inference with Markov Decision processes (MDPs).
- Formulated MPD to optimize genome sampling for estimation of growth rate parameter in an exponentially growing population under coalescent model.
- Used Python and bash to prepare genetic sequence data for analysis of pathogen phylogeny.
- Prepared medical image data for computer vision tasks (segmentation and classification).
- Completed two semesters of graduate coursework in genetics, statistics, and computer science.

Northern Arizona University *Flagstaff, AZ*

Research Technician *Aug. 2022 - Jul. 2023*

- Developed NIH-funded, Python-based open source epidemiological modeling software.
- Followed software engineering best practices by utilizing object-oriented design patterns, Git for version control, GitHub actions for CI/CD, and well-written documentation for project continuity.
- Mentored math and computer science undergraduate researchers.

Undergraduate Math Assistant *Aug. 2021 - May 2022*

- Studied the enumeration of signed permutations under the action of reversals and their applications to genome inversions.
- Solved several open problems, including finding a closed form solution for the number of signed permutations by reversal distance.
- Awarded $5,000 Hooper Undergraduate Research Award.
- Presented results at three different conferences.

Machine Learning Lab Research Assistant *Aug. 2021 - May 2022*

- Cleaned and implemented machine learning algorithms (Poisson regression, extreme gradient boosting, and random forests) on a large data set of multiplex polymerase chain reaction (mPCR) trials.
- Presented findings at undergraduate research symposium and published work to GitHub.

CUNY Baruch *Manhattan, NY (remote)*

Baruch Discrete Math REU Participant *Summer 2021*

- Adapted new methods for improving lower bound on Ramsey numbers to the multicolored bipartite setting.

Industry Experience

The CDC Foundation — *Atlanta, GA (Remote)*
Data Analyst — *Sep. 2024 - Present*
- Uses SQL, Python, and Tableau to extract, transform, load, and visualize non-profit fundraising data.
- Implements machine learning algorithms using Python (Pandas, Numpy, and Scikit-Learn) to aid in portfolio analysis and revenue forecasting.
- Automates time-consuming data quality and maintenance tasks using Python and Salesforce REST API.

ASM America — *Phoenix, AZ (Remote)*
Intern Data Analyst — *Summer 2022*
- Designed and built Power BI dashboards to visualize Global Product Team sales data

Teaching Experience

Northern Arizona University — *Flagstaff, AZ*
Calculus I Peer Teaching Assistant — *Aug. 2020 - Dec. 2020*
Linear Algebra Peer Teaching Assistant — *Jan. 2020 - May 2020*
Peer Math Assistant (Calculus I and III tutor) — *Jan. 2019 - May 2020*

Arizona Aspire Academy — *Phoenix, AZ*
Math and Music Tutor — *Aug. 2017 - May 2018*

Presentations

Posters
1. **Frank Burkhart**, Alexander Stewart, 2022. The Enumeration of signed permutations Under the Actions of Reversals. *Joint Mathematics Meeting*.
2. **Frank Burkhart**, Alexander Stewart, 2022. The Enumeration of signed permutations Under the Actions of Reversals. *Northern Arizona University Undergraduate Research Symposium*.
3. **Frank Burkhart**, Alexander Stewart, 2022. The Enumeration of signed permutations Under the Actions of Reversals. *Hooper Undergraduate Research Award Symposium.*
4. **Frank Burhart,** 2020. Machine Learning Algorithms for Multiplex Polymerase Chain Reaction Primer Design. *Northern Arizona University Undergraduate Research Symposium.*

Oral Presentations
1. **Frank Burkhart**, 2023**.** The Enumerative Combinatorics of Genome Inversions. *NC State Phylodynamics Research Group*.
2. **Frank Burkhart**, 2021. Bounds on Bipartite Multicolored Ramsey Numbers. *Baruch CUNY Discrete Math Group.*
3. **Frank Burkhart,** 2020. Using mlr3 for Machine Learning Research. *Northern Arizona University Machine Learning Lab*.

Awards and Honors
- NIEHS Environmental Health Bioinformatics (EHB) Fellowship — *Aug. 2023*
- NAU Department of Mathematics and Statistics Outstanding Senior — *May 2022*
- J. Harvey Butchart Mathematics Scholarship — *Jan. 2021*
- Hooper Undergraduate Research Award (HURA) — *Jan. 2021*

Technical Skills
Favorite Languages and Tools: Python, R, SQL, Tableau, Git, and LaTeX.